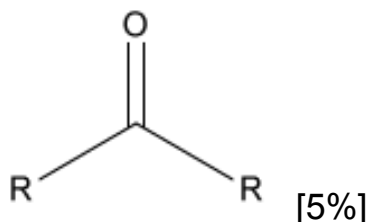
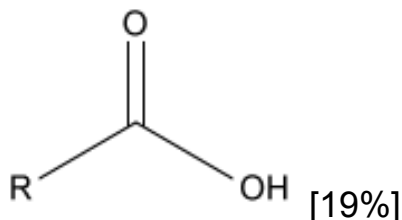


A carboxylic acid is reacted with NaOH in an organic solvent followed by a reaction with an acid chloride. What is the general structure of the carboxylic acid derivative formed in this reaction?

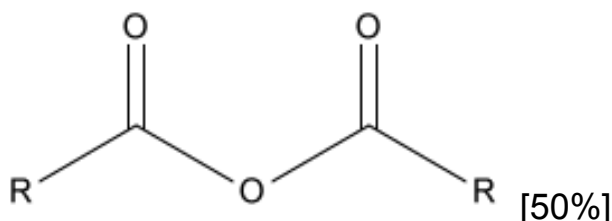
☐ A.



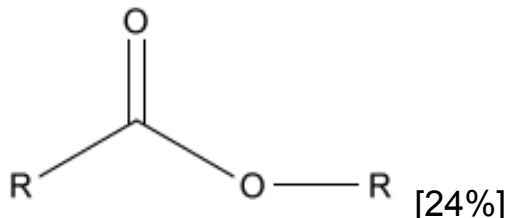
☐ B.



☒ C.



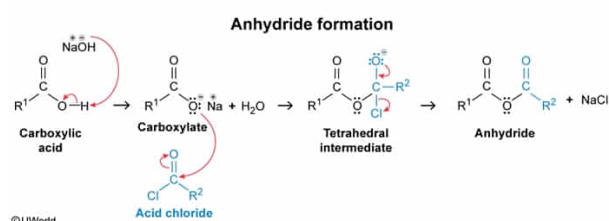
☐ D.



Correct

50% answered correctly

Explanation:



Anhydrides are **carboxylic acid derivatives** produced by condensation of **two carboxylic acids** or **nucleophilic acyl substitution** of an **acid chloride** with a **carboxylate**. The anhydride product is characterized by two carbonyl groups joined by an oxygen atom. An organic R group (typically, an alkyl group) is also bonded to each carbonyl carbon; these alkyl groups may be identical to each other or they may be distinct. Anhydride formation results in the **loss** of a **water** molecule from the carboxylic acid starting material.

When a carboxylic acid and NaOH react, hydroxide (OH^-) deprotonates the carboxylic acid, forming a carboxylate (carboxylic acid salt) and H_2O . When an acid chloride is added to the reaction, the negatively charged carboxylate acts as a nucleophile to attack the acid chloride carbonyl, yielding a tetrahedral intermediate. Chloride is then eliminated from the tetrahedral intermediate, and an anhydride is formed.

(Choice A) This is a ketone. Ketones are not a carboxylic acid derivative and therefore cannot form from an acid chloride and a carboxylate.

(Choice B) This is a carboxylic acid. It is a reagent in the reaction rather than a product.

(Choice D) This is an ester. **Esters** are made from an acid chloride and an alcohol or a carboxylic acid and an alcohol rather than a carboxylic acid and an acid chloride.

Educational objective:

Anhydrides are carboxylic acid derivatives characterized by two carbonyl groups joined by an oxygen atom. Anhydrides can be prepared by condensation of two carboxylic acid molecules or nucleophilic acyl substitution of an acid chloride by a carboxylate. Both reactions result in the loss of a water molecule from the carboxylic acid starting material.

Time spent:0 sec

Subject:

Organic Chemistry

QID:402440

System:

Functional Groups and Their Reactions